

Numerical and analytical study of the gravitational capture in the bicircular problem

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Abstract

The objective of the present paper is to study the ballistic gravitational capture in a dynamical model that has the presence of four bodies. In particular, the Earth–Moon–Sun–Spacecraft system is considered. This phenomenon is explained in terms of the integration of the perturbing forces with respect to time. Analytical equations are derived to estimate their magnitude and to show the best directions of approaching for the spacecraft. Using those equations, an analytical estimate of the fourth body effect valid for any system of primaries is derived. Numerical simulations are also performed to show the differences between the models with three and four bodies in the trajectories of the spacecraft.

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1. Introduction

A gravitational capture occurs when a spacecraft (or any particle with negligible mass) change from a hyperbolic orbit with a small positive energy around a celestial body into an elliptic orbit with a small negative energy without the use of any propulsive system. The force responsible for this modification in the orbit of the spacecraft is the gravitational force of the third and the fourth bodies involved in the dynamics. In this way, those forces are used as a zero cost control, equivalent to a continuous thrust applied in the spacecraft. One of the most important applications of this property is the construction of trajectories to the Moon. The concept of gravitational capture is used together with the basic ideas of the gravity-assisted maneuver and the bi-elliptic transfer orbit to generate

a trajectory that requires a fuel consumption smaller than the one required by the Hohmann transfer. This maneuver consists of the following steps: (i) the spacecraft is launched from an initial circular orbit with radius r_0 to an elliptic orbit that crosses the Moon's path; (ii) a Swing-By with the Moon is used to increase the apoapsis of the elliptic orbit. This step completes the first part of the bi-elliptic transfer, with some savings in ΔV due to the energy gained from the Swing-By; (iii) with the spacecraft at the apoapsis, a second very small impulse is applied to rise the periapsis to the Earth–Moon distance. The solar effects can reduce even more the magnitude of this impulse; (iv) the transfer is completed with the gravitational capture of the spacecraft by the Moon.

The application of this phenomenon in spacecraft trajectories is recent in the literature. The first demonstration of this was in Belbruno (1987). Further studies include Belbruno (1990, 1992), Krish (1991), Krish et al. (1992), Miller and Belbruno (1991) and Belbruno and Miller (1990, 1993). They all studied missions in

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the Earth–Moon system that use this technique to save fuel during the insertion of the spacecraft in its final orbit around the Moon. Another set of papers that made fundamental contributions in this field, also with the main objective of constructing real trajectories in the Earth–Moon system, are those of Yamakawa et al. (1992, 1993) and Yamakawa (1992). The first real application of a ballistic capture transfer was made during an emergency in a Japanese spacecraft (Belbruno and Miller, 1990). After that, some studies that consider the time required for this transfer appeared in the literature. Examples of this approach can be found in the papers by Vieira Neto and Prado (1995, 1998). An extension of the dynamical model to consider the effects of the eccentricity of the primaries is also available in the literature (Vieira Neto and Prado, 1996; Vieira Neto, 1999). A study of this problem, from the perspective of invariant manifolds, was developed by Belbruno (1994). Regarding applications of the bicircular problem, some of the important papers are: Simo et al. (1995), Gomez et al. (2001a), Gomez et al. (2001b) and Castella and Jorba (2000).

The present paper has the goal of developing analytical equations to estimate the effects of the fourth body in the gravitational capture. The main developments are made for the Earth–Moon–Sun–Spacecraft system, but many results are valid for any system of primaries.

2. Mathematical models

In the bicircular problem (Cronin et al., 1964), we have (Fig. 1): (i) the Earth and the Moon constitute the two main primaries of the system, both in circular orbits around the common center of mass; (ii) the Sun is the third body, assumed to be in a circular orbit around the center of mass of the Earth–Moon system and its orbit is coplanar with the orbit of the Moon around the Earth; (iii) the trajectory of a massless fourth body that has its motion governed by the three primaries

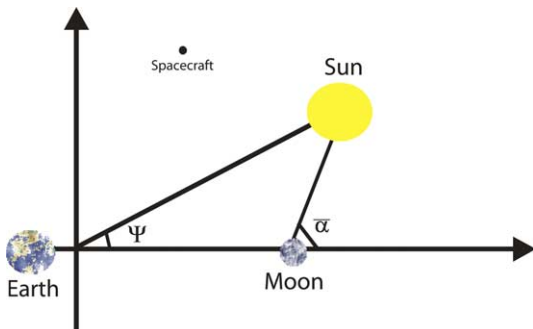


Fig. 1. Restricted four-body model (Cartesian coordinate).

is studied. The center of mass of the Earth–Moon system is the origin of the reference system. The positions of the bodies are: $(-\mu_2, 0, 0)$ for the Earth and $(\mu_1, 0, 0)$ for the Moon. The values of the masses are given by $\mu_1 = 0.9878493317$ for the Earth, $\mu_2 = 0.0121506683$ for the Moon and $\mu_{\text{sun}} = 328900.48$ for the Sun (canonical units). To obtain the equations of motion, the same procedure used by Yamakawa (1992) is used here. The Lagrangian is written as:

$$L = T - U,$$

where T is the kinetic energy given by

$$T = \frac{1}{2} \left(\left(\frac{dx}{dt} \right)^2 + \left(\frac{dy}{dt} \right)^2 + \left(\frac{dz}{dt} \right)^2 \right) + \left(x \frac{dy}{dt} - \frac{dx}{dt} y \right) + \frac{1}{2} (x^2 + y^2)$$

and U is the potential energy given by

$$U = -\frac{\mu_1}{r_1} - \frac{\mu_2}{r_2} - \mu_{\text{sun}} \left[\frac{1}{r_{\text{sun}}} - \frac{xx_s + yy_s}{R_{\text{sun}}^3} \right],$$

where r_1 , r_2 , and r_{sun} represents the distances between the spacecraft and the Earth, the Moon and the Sun, respectively. $R_{\text{sun}} = 389.1723985$ is the distance between the Sun and the origin of the coordinate system. (x_s, y_s, z_s) represents the position of the Sun with respect to the rotating system. Those variables are given by:

$$r_1^2 = (x + \mu_2)^2 + y^2 + z^2,$$

$$r_2^2 = (x - \mu_1)^2 + y^2 + z^2,$$

$$r_{\text{sun}}^2 = (x - x_s)^2 + (y - y_s)^2 + (z - z_s)^2 \\ = (x + R_{\text{sun}} \cos \psi)^2 + (y - R_{\text{sun}} \sin \psi)^2 + z^2,$$

$$\psi = \bar{\alpha} + (\omega_{\text{E-M}} - \Omega_{\text{sun}})t,$$

where $\bar{\alpha}$ is the phase angle of the Sun, given by (anti-Sun direction)–(origin)–(Moon), that is the initial position of the Moon in a system that has the Earth and the Sun in fixed positions; $\omega_{\text{E-M}}$ is the angular velocity of the Earth–Moon system ($=1.00$); Ω_{sun} is the angular velocity of the Sun with respect to the system of reference ($=0.07480133$).

From the Lagrangian, it is possible to obtain the equations of motion, that are given by:

$$\begin{aligned} \frac{d^2x}{dt^2} - 2 \frac{dy}{dt} - x - \frac{\mu_{\text{sun}}}{R_{\text{sun}}^2} \cos \psi &= -\frac{\partial U_n}{\partial x}, \\ \frac{d^2y}{dt^2} + 2 \frac{dx}{dt} - y + \frac{\mu_{\text{sun}}}{R_{\text{sun}}^2} \sin \psi &= -\frac{\partial U_n}{\partial y}, \\ \frac{d^2z}{dt^2} &= -\frac{\partial U_n}{\partial z}, \end{aligned} \quad (1)$$

where

$$U_n = -\frac{\mu_1}{r_1} - \frac{\mu_2}{r_2} - \frac{\mu_{\text{sun}}}{r_{\text{sun}}},$$

$$\frac{\partial U_n}{\partial x} = \frac{\mu_1}{r_1^3}(x + \mu_2) + \frac{\mu_2}{r_2^3}(x - \mu_1) + \frac{\mu_{\text{sun}}}{r_{\text{sun}}^3}(x + R_{\text{sun}} \cos \psi),$$

$$\frac{\partial U_n}{\partial y} = \frac{\mu_1}{r_1^3}(y) + \frac{\mu_2}{r_2^3}(y) + \frac{\mu_{\text{sun}}}{r_{\text{sun}}^3}(y - R_{\text{sun}} \cos \psi),$$

$$\frac{\partial U_n}{\partial z} = \frac{\mu_1}{r_1^3}(z) + \frac{\mu_2}{r_2^3}(z) + \frac{\mu_{\text{sun}}}{r_{\text{sun}}^3}(z).$$

The standard canonical system of units is used, in which: the unit of distance is the distance between M_1 (the Earth) and M_2 (Moon); the angular velocity (ω_{E-M}) of the motion of M_1 and M_2 is set to unity; the mass of the smaller primary (M_2) is given by $\mu = m_2/(m_1 + m_2)$ (where m_1 and m_2 are the real masses of the Earth and the Moon, respectively) and the mass of M_2 is $(1 - \mu)$; the unit of time is defined such that the period of the motion of the two primaries is 2π and, as a consequence of those choices, the gravitational constant is unity.

When considering the restricted three-body problem, the well known equations shown below are used, also in the canonical system of units:

$$\ddot{x} - 2\dot{y} = \frac{\partial \Omega}{\partial x}, \quad \ddot{y} + 2\dot{x} = \frac{\partial \Omega}{\partial y},$$

where Ω is the pseudo-potential given by $\Omega = \frac{1}{2}(x^2 + y^2) + \frac{(1-\mu)}{r_1} + \frac{\mu}{r_2}$.

3. Forces involved in the dynamics

The criterion to consider a trajectory as a “ballistic gravitational capture” is to verify if the distance from the Moon reaches 100,000 km in a time smaller than 50 days without the use of any propulsive system (Yamakawa, 1992). A sphere centered on the Moon with radius of 100,000 km will be called here the “gravitational sphere of capture” of the Moon. Then, the main parameters to identify one trajectory are: r_p , the perapsis distance of the incoming trajectory, that is the closest point that the spacecraft reaches with respect to the Moon before receiving an impulse that completes the capture (assumed to be 1838 km in the calculations performed in the present paper); α , the perapsis position angle, that is the angle between the line that connects the Earth and the Moon and the line that connects the Moon with the perapsis point (shown in Figs. 2 and 3) and β , the entry position angle, that is the angle that specifies the point where the spacecraft reaches the sphere of capture of the Moon (angle between the Earth–Moon line and the line linking the Moon with the point where the spacecraft reaches the sphere of influence of the Moon).

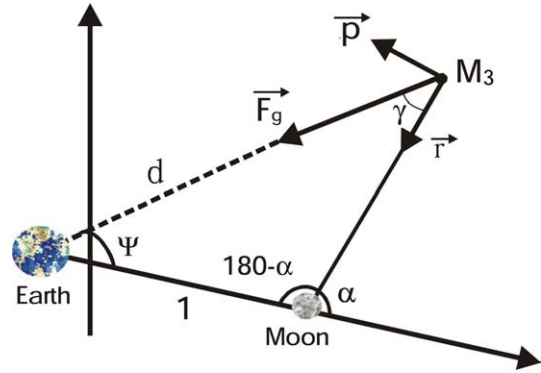


Fig. 2. Gravitational force of the Earth.

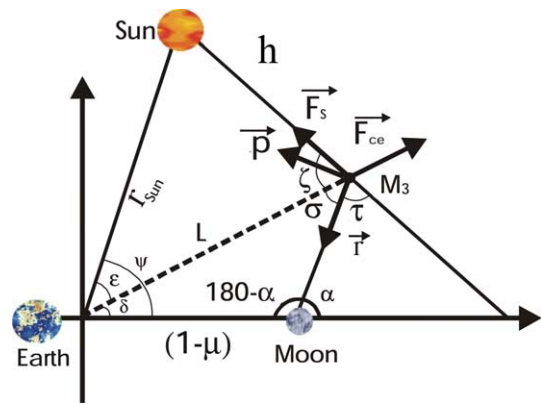


Fig. 3. The gravitational force of the Sun and the centrifugal force.

To understand better the physical reasons of this phenomenon, it is useful to calculate the forces acting over the massless particle. Fig. 2 shows the gravitational force $\vec{F}_g$ of the Earth acting in a spacecraft M_3 that is approaching the Moon and Fig. 3 shows the gravitational force of the Sun ($\vec{F}_s$) and the centrifugal force ($\vec{F}_{ce}$) acting in the same situation. There is also the Coriolis force, but this force is not analyzed in detail because the main idea of this paper is to explain the ballistic gravitational capture as a result of perturbative forces acting in this direction and the Coriolis force acts perpendicular to the direction of motion of the spacecraft all the time. The direction $\vec{r}$ points directly to the center of the Moon and the direction $\vec{p}$ is perpendicular to $\vec{r}$, pointing in the counter-clockwise direction. The distance between the spacecraft and the Earth is d , the angle formed by the line connecting the Earth to the spacecraft and the direction $\vec{r}$ is γ . The angle ϕ is used to define instantaneously the direction $\vec{r}$, because it is the angle that the radial trajectory with the Earth–Moon line. Under the assumption of radial trajectories, $\phi = \alpha$ in all situations. From geometrical considerations shown in more detail in Prado (2002), it is possible to write for the gravitational force

$$\begin{aligned}\vec{F}_g &= \frac{(1-\mu)(r+\cos\phi)}{d^3}\vec{r} + \frac{(1-\mu)\sin\phi}{d}\vec{p} \\ &= \frac{(1-\mu)(r+\cos\phi)}{(1+r^2+2r\cos\phi)^{3/2}}\vec{r} + \frac{(1-\mu)\sin\phi}{(1+r^2+2r\cos\phi)^{1/2}}\vec{p}.\end{aligned}\quad (2)$$

For the centrifugal force the expression is (Prado, 2002)

$$\vec{F}_{ce} = -[r + (1-\mu)\cos\phi]\vec{r} + (\mu-1)\sin\phi\vec{p}. \quad (3)$$

In the present paper, the contribution for the analytical study of this problem is to develop an equation for the gravitational force of the Sun ($\vec{F}_s$) and its effects in the gravitational capture, to complete the results available in the literature for the gravitational force of the Earth and the centrifugal force. From Fig. 3, it is possible to find the following relations, where $\vec{F}_{Sr}$ stands for the radial component and $\vec{F}_{Sp}$ stands for the perpendicular component:

$$\vec{F}_{Sr} = -\frac{\mu_{\text{sun}}\cos\tau}{h^2}\vec{r}, \quad \vec{F}_{Sp} = -\frac{\mu_{\text{sun}}\sin\tau}{h^2}\vec{p}, \quad (4)$$

$$R_{\text{sun}}^2 = L^2 + h^2 - 2Lh\cos\zeta \Rightarrow \cos\zeta = \frac{R_{\text{sun}}^2 - L^2 - h^2}{-2Lh}, \quad (5)$$

$$\tau = \pi - \sigma - \zeta, \quad (6)$$

$$\delta = \alpha - \sigma, \quad (7)$$

$$\varepsilon = \psi - \delta = \psi - \alpha + \sigma, \quad (8)$$

$$z^2 = L^2 + R_{\text{sun}}^2 - 2LR_{\text{sun}}\cos\varepsilon. \quad (9)$$

These angles are shown in Figs. 2 and 3. σ is the angle between the radial trajectory and the line that links the spacecraft to the origin of the reference system and ζ is the angle between the line that links the spacecraft to the origin of the reference system and the line that links the spacecraft to the Sun.

The analytical equations to measure the effects of this perturbation in the gravitational capture are derived under the assumption that the trajectory followed by the spacecraft is an idealized trajectory that does

not deviate from the radial direction. The real trajectories are not radial, but the equations derived under this assumption can be used to study the effect of the fourth body in the savings obtained in the gravitational capture. Another justification for the radial trajectories used to derive the equations is that the reduction of C_3 (that is twice the value of the two-body energy of the Moon–Spacecraft system) is a result of the effects of the forces in time during the whole trajectory and, even for trajectories that shows several loops before arriving at the periapsis, during most of the time the trajectory can be seen as composed by a set of trajectories close to radial.

For the derivation performed here, the component measured is the radial one, because this is the direction of motion under the assumption made in the present research. Then, assuming that the spacecraft is in free-fall (subject only to the gravitational and centrifugal forces) traveling with zero energy (parabolic trajectory) and that the trajectories do not deviate from a straight line, the result is

$$\text{Total energy} = E = 0 = \frac{1}{2}V^2 - \frac{\mu}{r} \Rightarrow V = \sqrt{\frac{2\mu}{r}} = \frac{ds}{dt}, \quad (10)$$

where ds is the space traveled by the particle during the time dt . To obtain the integral of the effect of the perturbing forces with respect to time, it is possible to perform the calculations in terms of the radial distance, by making the substitution

$$\int_{t_0}^{t_f} F dt = \int_{S_0}^{S_f} (F/V) ds = \int_{r_{\min}}^{r_{\max}} (F/V) dr. \quad (11)$$

The extreme points of the integration changes position (S_0 by $r_{\min}$ and S_f by $r_{\max}$) to take into account that the positive sense of the radial direction points towards the Moon. Since the spacecraft is assumed to approach the Moon on a radial trajectory the result $\phi = \alpha = \beta$ is valid, and the variable is used as the independent parameter. Remember that $q = (r_{\max} + r_{\min})/2$ is the middle point of the trajectory. Performing this process for the gravitational force due to the Sun, we have:

$$\begin{aligned}F_{\text{sun}}(\alpha) &= \left[\left(\frac{\mu_{\text{sun}}(q + \cos\alpha - \mu\cos\alpha - r_{\text{sun}}\cos(\alpha - \psi))}{\sqrt{2\mu/q}(1 + q^2 + r_{\text{sun}}^2 - 2\mu + \mu^2 + 2q(1-\mu)\cos\alpha - 2qr_{\text{sun}}\cos(\alpha - \psi) - 2r_{\text{sun}}\cos\psi + 2r_{\text{sun}}\mu\cos\psi)^{3/2}} \right) r \right]_{r_{\min}}^{r_{\max}} \\ &+ \left[\left(-\frac{3(2q + 2(1-\mu)\cos\alpha - 2r_{\text{sun}}\cos(\alpha - \psi))(q + \cos\alpha - \mu\cos\alpha - r_{\text{sun}}\cos(\alpha - \psi))}{\sqrt{2\mu/q}(1 + q^2 + r_{\text{sun}}^2 - 2\mu + \mu^2 + 2q(1-\mu)\cos\alpha - 2qr_{\text{sun}}\cos(\alpha - \psi) - 2r_{\text{sun}}\cos\psi + 2r_{\text{sun}}\mu\cos\psi)^{5/2}} \right. \right. \\ &\left. \left. + \frac{\sqrt{q/\mu} + \sqrt{q/\mu} \left(\frac{(q + \cos\alpha - \mu\cos\alpha - r_{\text{sun}}\cos(\alpha - \psi))}{2q} \right)}{(1 + q^2 + r_{\text{sun}}^2 - 2\mu + \mu^2 + 2q(1-\mu)\cos\alpha - 2qr_{\text{sun}}\cos(\alpha - \psi) - 2r_{\text{sun}}\cos\psi + 2r_{\text{sun}}\mu\cos\psi)^{3/2}} \right) \left(\frac{\mu_{\text{sun}}}{\sqrt{2}} \right) \left(\frac{r^2}{2} - qr \right) \right]_{r_{\min}}^{r_{\max}}.\end{aligned}\quad (12)$$

This equation is valid for any system of primaries, if we understand that μ_{sun} is the gravitational parameter of the fourth body and r_{sun} is the distance between the spacecraft and the fourth body. Using the numerical values for the Earth–Moon–Sun–Spacecraft system ($\mu = 0.0121$, $\mu_{\text{sun}} = 328900.48$, $R_{\text{sun}} = 389.1723985$, $r_{\text{max}} = 100000/384400$, $r_{\text{min}} = 1838/384400$, $q = (r_{\text{max}} + r_{\text{min}})/2$), we have that the effect of the fourth body is only a function of the angles α and ψ . This equation is

$$F_{\text{sun}} = \frac{196501(0.1325 + 0.9879 \cos(\alpha) - 389.1720 \cos(\alpha - \psi))}{(151456 + 0.2617 \cos(\alpha) - 103.1020 \cos(\alpha - \psi) - 768.9270 \cos(\psi))^{3/2}}. \quad (13)$$

The second terms have a small contribution, compared with the other forces, to be shown here. Fig. 4 shows the effects of the gravitational force of the Sun as a function of α and ψ , using Eq. (13). It is clear that the best result, regarding capture, occurs for $\alpha = \psi$, that means that the spacecraft is aligned with the Sun, what is an expected result. After making $\alpha = \psi$, it is possible to make simulations and they show that the value of $\alpha = \psi = 0$ gives the maximum effect for the fourth body perturbation. The analytical derivations performed in this research were made using the software Mathematica.

For the radial component of the Earth's gravity, up to the first order, the integral is already known and it is given by (Prado, 2002):

$$F_1(\alpha) = \left[\frac{(1-\mu)(q+\cos\alpha)}{(2\mu/q)^{1/2}(1+q^2+2q\cos\alpha)^{3/2}} r + (1-\mu) \left(-\frac{3(q+\cos\alpha)^2}{(2\mu/q)^{1/2}(1+q^2+2q\cos\alpha)^{5/2}} + \frac{\mu(3q+\cos\alpha)}{q^2(2\mu/q)^{3/2}(1+q^2+2q\cos\alpha)^{3/2}} \right) \left(\frac{r^2}{2} - qr \right) \right]_{r_{\text{min}}}^{r_{\text{max}}}. \quad (14)$$

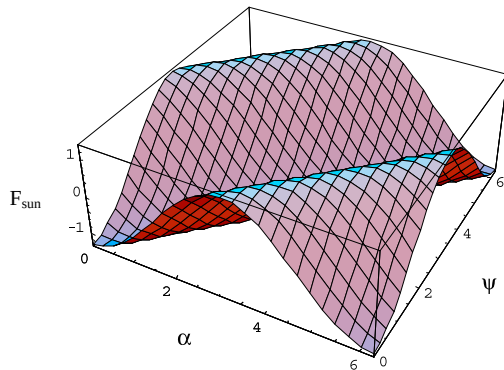


Fig. 4. Effects of the gravitational force of the Sun as a function of α and ψ .

For the radial component of the centrifugal force, the integral is also available in the literature (Prado, 2002)

$$\begin{aligned} & \int_{r_{\text{min}}}^{r_{\text{max}}} (F_{\text{ce}}/V) ds \\ &= \int_{r_{\text{min}}}^{r_{\text{max}}} ((\mu-1)\cos\alpha + r)(2\mu/r)^{-1/2} dr \\ &= \left[\left(-0.4r^2 + \frac{2}{3}r(\mu-1)\cos\alpha \right) \times (2\mu/r)^{-1/2} \right]_{r_{\text{min}}}^{r_{\text{max}}}. \quad (15) \end{aligned}$$

4. Estimating the effects of the fourth-body for different systems of primaries

The analytical equations obtained above can be used to estimate the effects of the fourth-body as a function of the mass parameter μ , which identifies the system of primaries, and the parameters of the fourth body μ_{4B} (gravitational) and R_{4B} (distance from the center of the system). To perform this estimate, the distance of the sphere of capture (r_{max}) and the periapsis distance (r_{min}) of the trajectory are given by (Yamakawa, 1992) $(2\mu)^{1/3}$ and $(2\mu)^{1/3}/20$, respectively. Consider also that there is an ideal trajectory that is radial and traveling with $\alpha = \psi = 0^\circ$, that is the radial direction that provides the maximum effect of the fourth body. The equation that results from those assumptions is a function only of the mass parameter μ and the parameters of the fourth body μ_{4B} and R_{4B} . This final equation is

$$F_{4B} = \frac{76\sqrt{210}\mu_{4B}}{(40 - 40R_{4B} + (212\mu)^{1/3} - 40\mu)^2}. \quad (16)$$

5. Numerical simulations

To show better the differences between the dynamical systems including three and four bodies, Fig. 5 has plots of two trajectories with the same initial conditions ($\psi = 0$, $\alpha = 180^\circ$, $C_3 = -0.18$) and different dynamics. The fine line represents the trajectory obtained using the bicircular four-body dynamical system and the darker line represents the trajectory obtained using the restricted three-body problem. Note that the trajectories are very similar in the beginning of the time and then they split into two different trajectories. The Sun has the effect of avoiding the loop that occur under the restricted three-body model, what makes the trajectory

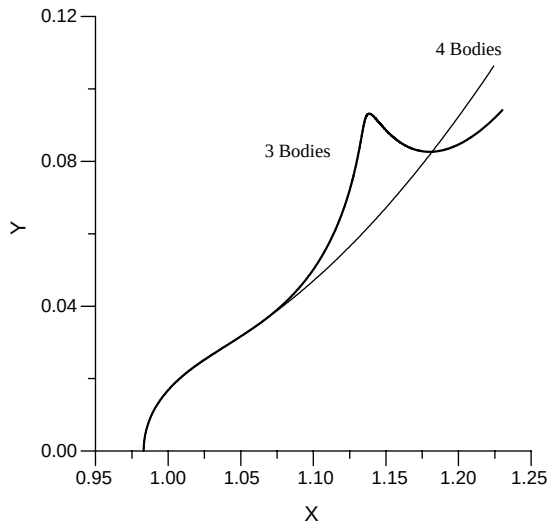
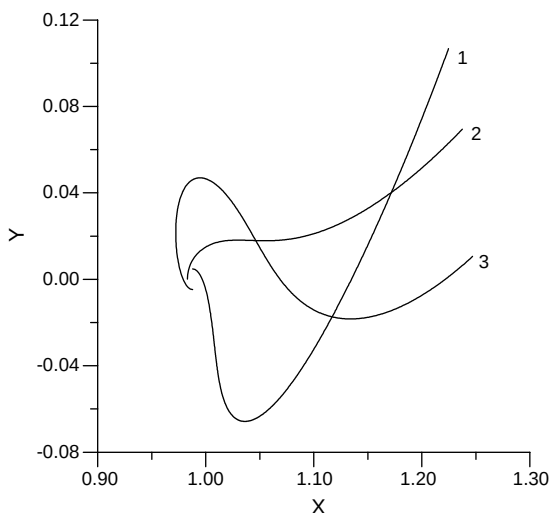
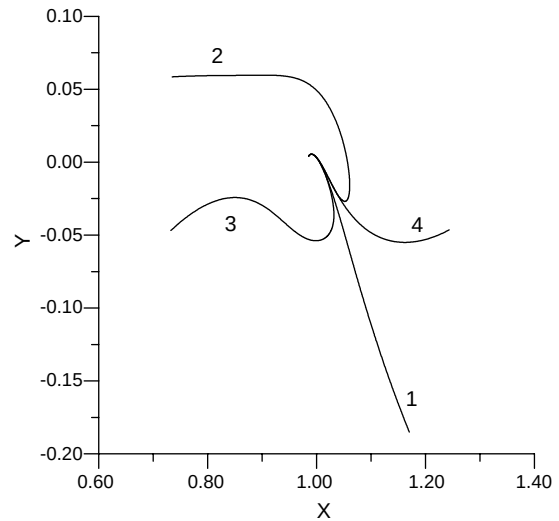


Fig. 5. Trajectories obtained under the two dynamics considered.

close to a single line. The numerical integration performed here and in all phases of the present research used the Runge–Kutta 7–8 method.

Then, this paper studies the effects of the two main parameters of the system: α and ψ . First, the values for the angle ψ is fixed in zero, C_3 in -0.4 and α is assumed to have the values 90° (trajectory 1), 180° (trajectory 2), 270° (trajectory 3). The trajectories are shown in Fig. 6. They have different characteristics and time to reach gravitational capture. This quantitative study shows that the effect of the Sun is to create a preferential direction of motion for the trajectories. The three trajectories start in different positions, but they all go to the right side of the plots, that is where the Sun is located (remember that $\psi = 0$). The fixed value of $C_3 = -0.4$ is also an indication of the effect of the Sun. Under the model given by the restricted three-body problem, the

Fig. 6. Trajectories for different values of α .Fig. 7. Trajectories for different values of ψ .

minimum value that C_3 can reach is above -0.25 . Including the presence of the Sun, the value of -0.40 can be reached. The angle ψ gives the position of the Sun at a given instant of time, so it has a very important rule in this study. Depending on this value, the Sun can be located in a position that can accelerate the spacecraft, situation where the savings obtained are smaller when compared to the results obtained under the model given by the restricted three-body problem, or breaking the spacecraft, which means increasing the savings obtained by the model with three bodies. There are some specific positions that can allow savings a lot larger than the ones available under the model given by the restricted three-body problem.

To verify the influence of the angle ψ four more simulations are made. The value of C_3 is fixed in -0.15 , α in 120° and ψ is varied in the values 0° , 90° , 180° and 270° . Fig. 7 shows the four trajectories. The trajectories show that the spacecraft start in similar paths when close to the Moon, because the influence of the Sun is not very strong at this point, and then the trajectories deviate from each other, going to the direction of the Sun. It shows that the Sun start to have a very large effects in the trajectories when the spacecraft escape from the Moon.

6. Conclusions

This paper had the main goal of studying the ballistic gravitational capture problem under the model given by the bicircular problem. It showed an explanation of the phenomenon based in the calculation of the forces involved in the dynamics as a function of time and in its integration with respect to time. It also derived analytical equations to study the

effect of the fourth body, under the assumption of radial motion. There are three forces that act as disturbing forces in the direction of motion: the gravitational forces due to the Earth and the Sun and the centrifugal force. These forces can slow down the motion of the spacecraft, working opposite to its motion. This is equivalent to applying a continuous propulsion force against the motion of the spacecraft. Then, the equations are generalized to be used in any system of primaries. The numerical results showed that large savings can be obtained under the bicircular four body problem when compared to the restricted three body problem. In order to obtain those savings, it is necessary to find a proper geometry to start the maneuver, such that the force of the Sun acts to reduce the velocity of the spacecraft that is approaching the Moon. If this is not done, the savings can be reduced or even disappear. After that, a numerical simulation of two trajectories with the same initial conditions under two different dynamics shows the effect of the fourth body.

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